

Midterm 1 on Monday, good luck

Per request of classroom observers from 2 weeks ago, different format for just this week: review problems 2 by 2, 12 minutes per pair, 1&2, then 3&4, ..., then 9&10.

Instructors plan on requiring Poll Everywhere for end of class attendance starting next week.

6: $r(t) = \langle t, t^2, 4 \rangle$, find T, N, K

(a) $r'(t) = \langle 1, 2t, 0 \rangle$, $|r'| = \sqrt{1+4t^2}$

So $T(t) = \frac{r'(t)}{\|r'(t)\|} = \frac{1}{\sqrt{1+4t^2}} \langle 1, 2t, 0 \rangle$

or $\left\langle \frac{1}{\sqrt{1+4t^2}}, \frac{2t}{\sqrt{1+4t^2}}, 0 \right\rangle$

(b) Recall $N(t) = \frac{T'(t)}{\|T'(t)\|}$, so we

need to find T' first. Also recall,

$$\frac{d}{dt}(f\vec{v}) = f'\vec{v} + f\vec{v}'. \quad \text{So}$$

$$\begin{aligned}
T' &= \left[(1+4t^2)^{1/2} \right] \langle 1, 2t, 0 \rangle + \frac{1}{\sqrt{1+4t^2}} \langle 0, 2, 0 \rangle \\
&= \frac{8t + (-\frac{1}{2})}{(1+4t^2)^{3/2}} \langle 1, 2t, 0 \rangle + \frac{1}{\sqrt{1+4t^2}} \langle 0, 2, 0 \rangle \\
&= \frac{-4t \langle 1, 2t, 0 \rangle + (1+4t^2) \langle 0, 2, 0 \rangle}{(1+4t^2)^{3/2}} \\
&= \frac{\langle -4t, 2, 0 \rangle}{(1+4t^2)^{3/2}} = \frac{2}{(1+4t^2)^{3/2}} \langle -2t, 1, 0 \rangle.
\end{aligned}$$

Notice that for an expression of the form $f\vec{v}$, we have

$$\frac{f\vec{v}}{\|f\vec{v}\|} = \frac{f\vec{v}}{|f|\|\vec{v}\|} = \left(\frac{f}{|f|}\right) \frac{\vec{v}}{\|\vec{v}\|} = \frac{\vec{v}}{\|\vec{v}\|}$$

as long as $f \geq 0$, so

$$N = \frac{T'}{\|T'\|} = \frac{\langle -2t, 1, 0 \rangle}{\|\langle -2t, 1, 0 \rangle\|} = \frac{\langle -2t, 1, 0 \rangle}{\sqrt{4t^2 + 1}}$$

$$\text{Also } N = \frac{\cancel{r} \langle \dots \rangle}{|\cancel{r} \langle \dots \rangle|} = \frac{\cancel{r} \langle \dots \rangle}{\cancel{r} \sqrt{\dots}}$$

$$= \frac{\langle \dots \rangle}{\sqrt{\dots}} \text{ if you want to be}$$

extra sure no steps are skipped.

(c) Recall: $K = \frac{|T'(t)|}{|r'(t)|}$, and we

have already found both terms, so

just substitute:

$$K = \frac{\frac{2}{(1+4t^2)^{3/2}} |\langle -2t, 1, 0 \rangle|}{(1+4t^2)^{1/2}} = \frac{\frac{2(1+4t^2)^{1/2}}{(1+4t^2)^{3/2}}}{(1+4t^2)^{1/2}}$$

$$= 2(1+4t^2)^{-3/2}$$

4: Length of $r = \langle 2t, t^2, \frac{1}{3}t^3 \rangle$
with $0 \leq t \leq 1$.

Recall: $L = \int_a^b |r'(t)| dt$ for
a curve $a \leq t \leq b$ described by r .

$$r' = \langle 2, 2t, t^2 \rangle \Rightarrow |r'| = \sqrt{t^4 + 4t^2 + 4},$$

but square roots of ≥ 3 terms
are almost always impossible / difficult
to integrate, so look for a simplification.

Notice $t^4 + 4t^2 + 4 = (t^2 + 2)^2$, so

$$|r'| = t^2 + 2, \text{ so}$$

$$L = \int_0^1 (t^2 + 2) dt = \left. \frac{t^3}{3} + 2t \right|_0^1 = \frac{7}{3}.$$

5: $r(t) = \langle 3 \sin t, 4t, 3 \cos t \rangle$ describes
particle, starts @ $(0, 0, 3)$, find position
after particle travels 5π units.

Notice $(0, 0, 3) = r(0)$, so the particle starts at $t=0$.

Suppose it travels 5π units by $t=c$. If we can figure out c , then the position is just $r(c)$.

The arc length from $0 \rightarrow c$ must be 5π since it measures distance,

$$\text{so } 5\pi = \int_0^c |r'(t)| dt = \int_0^c 5 dt =$$

$$5c, \text{ so } c = \pi.$$

$$\begin{aligned} r'(t) &= \langle 3\cos t, 4, -3\sin t \rangle \\ |r'| &= \sqrt{4^2 + 9\cos^2 t + 9\sin^2 t} \\ &= \sqrt{16 + 9} = 5 \end{aligned}$$

So the answer is

$$r(\pi) = \langle 3\sin \pi, 4\pi, 3\cos \pi \rangle = \langle 0, 4\pi, -3 \rangle.$$

1: Derive $r = \langle e^{t^2+2t}, \ln(\cos t), t \arctan(t) \rangle$

$$(e^{t^2+2t})' = (t^2+2t)' e^{t^2+2t} = (2t+2)e^{t^2+2t}$$

$$\ln(\cos t)' = \frac{(\cos t)'}{\cos t} = -\frac{\sin t}{\cos t} = -\tan(t)$$

$$\begin{aligned} [t \arctan(t)]' &= \tan^{-1}t + t[\arctan(t)]' \\ &= \tan^{-1}t + \frac{t}{t^2+1} \end{aligned}$$

$$r' = \langle (2t+2)e^{t^2+2t}, -\tan(t), \tan^{-1}t + \frac{t}{t^2+1} \rangle$$

2: Find $\int_0^1 \langle \frac{1}{t+1}, \frac{1}{t^2+1}, \frac{t}{t^2+1} \rangle dt$

$$= \langle \int_0^1 \frac{dt}{t+1}, \int_0^1 \frac{dt}{t^2+1}, \int_0^1 \frac{t}{t^2+1} dt \rangle$$

$$= \langle \ln|t+1| \Big|_0^1, \arctan(t) \Big|_0^1, \frac{1}{2} \ln(t^2+1) \Big|_0^1 \rangle$$

$$= \langle \ln 2, \tan^{-1}(1), \frac{1}{2} \ln 2 \rangle = \langle \ln 2, \frac{\pi}{4}, \frac{1}{2} \ln 2 \rangle$$

3: Tangent line to $r = \langle t \cos t, t, \sin t \rangle$

at the point corr. to $t = \frac{\pi}{4}$.

Just like in Calc I, Slope of line is derivative at the point, so direction of the line will be $r'(\frac{\pi}{4})$. But also, the line passes through $r(\frac{\pi}{4})$ since it is tangent to the curve at $\frac{\pi}{4}$.

So the line is given by

$$r(\frac{\pi}{4}) + t r'(\frac{\pi}{4}).$$

$$r(\frac{\pi}{4}) = (\frac{\pi}{4} \cos \frac{\pi}{2}, \pi, \sin \frac{\pi}{4}) = \langle 0, \pi, \frac{1}{\sqrt{2}} \rangle$$

$$r'(t) = \langle \cos 2t - 2t \sin 2t, 4, \cos t \rangle$$

$$r'(\frac{\pi}{4}) = \langle -\frac{\pi}{2} \sin \frac{\pi}{2}, 4, \cos \frac{\pi}{4} \rangle = \langle -\frac{\pi}{2}, 4, \frac{\sqrt{2}}{2} \rangle$$

So final line equation is

$$\langle -\frac{\pi}{2}t, \pi + 4t, \frac{1}{\sqrt{2}} + t \frac{\sqrt{2}}{2} \rangle.$$

$$8: \quad r = \langle \ln t, t, e^{-t} \rangle$$

Just recall formulas for v, s, a .

$$v = r' = \langle \ln t + 1, 1, -e^{-t} \rangle$$

$$s = |v| = \sqrt{e^{-2t} + 1 + (\ln t + 1)^2}$$

$$a = v' = \langle \frac{1}{t}, 0, e^{-t} \rangle$$